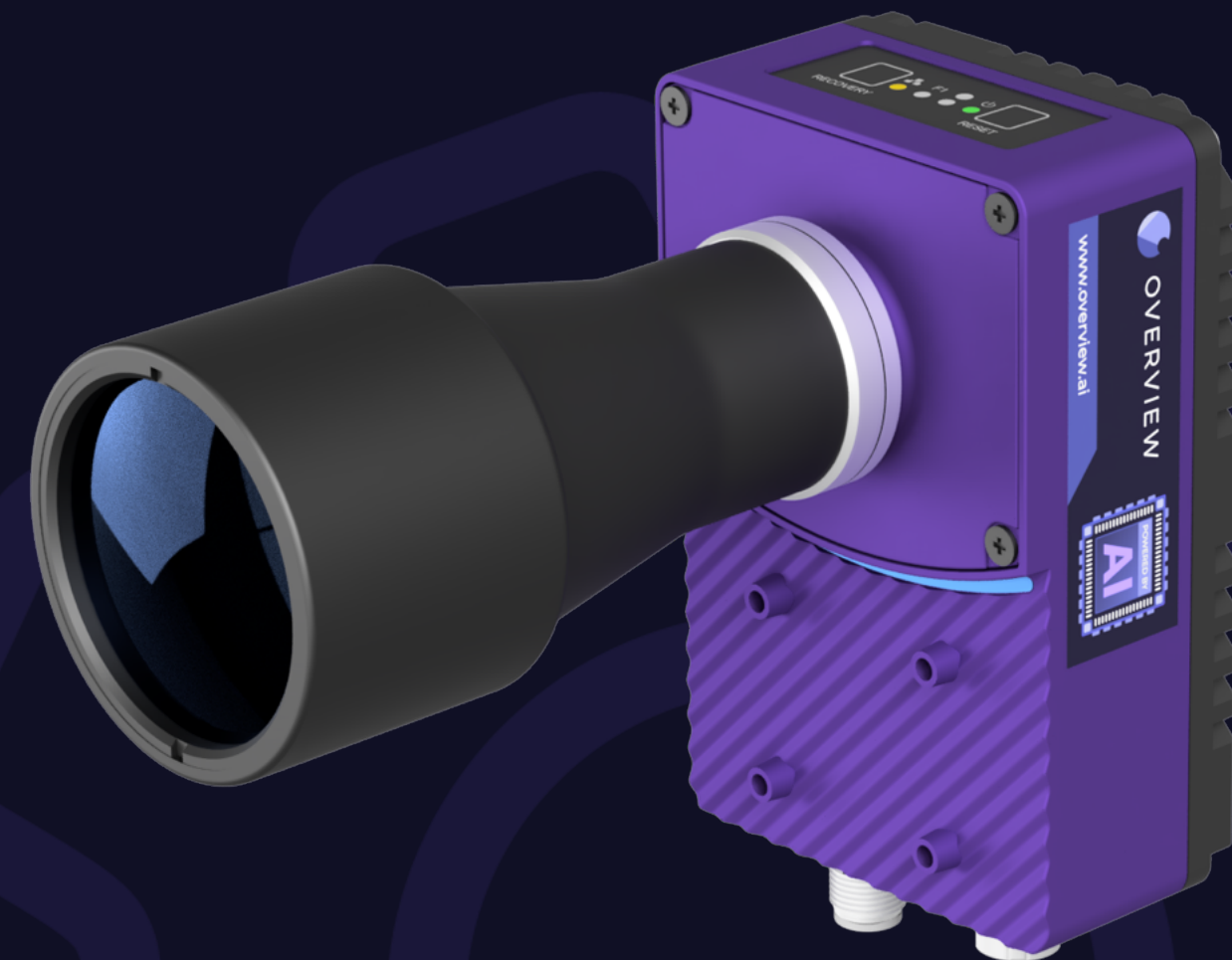


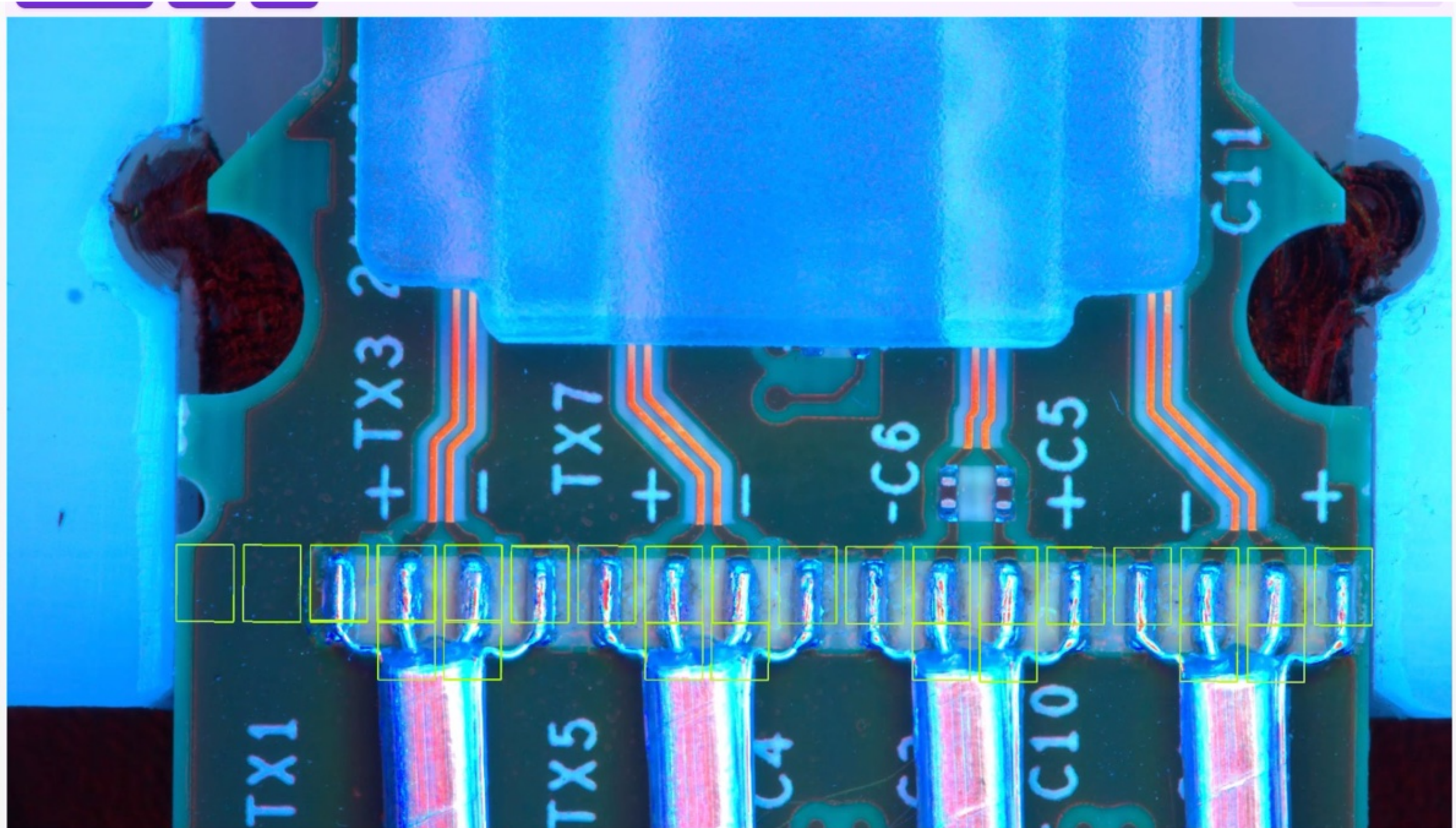
Hot Bar Soldering

OV80i Introduction



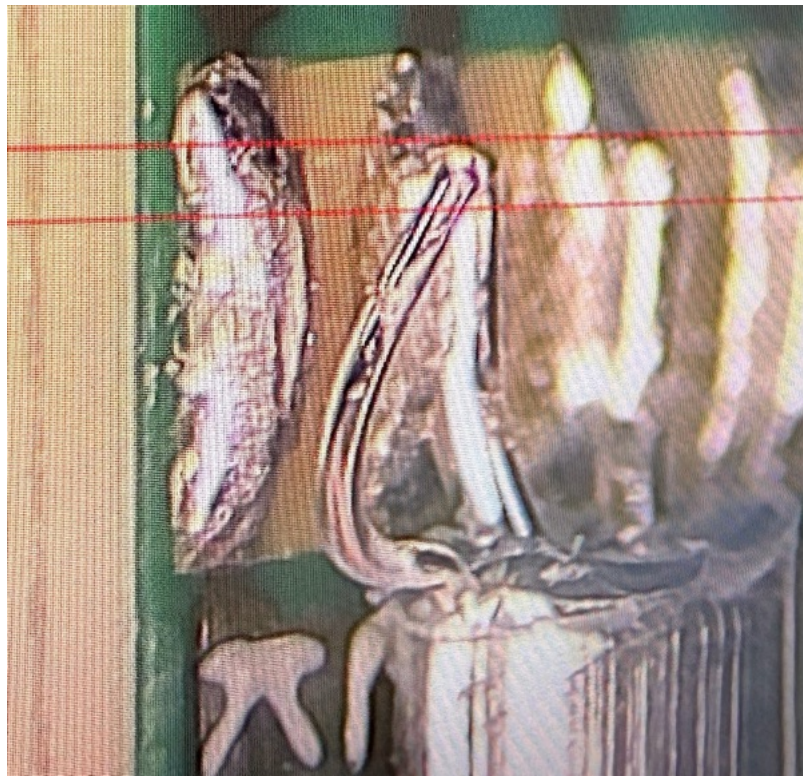
Hot Bar Soldering AI Inspection with OV80i

Successfully identified solder quality (including Z-axis defects), wire position, excess material.



Catching Z-Axis Solder Defects

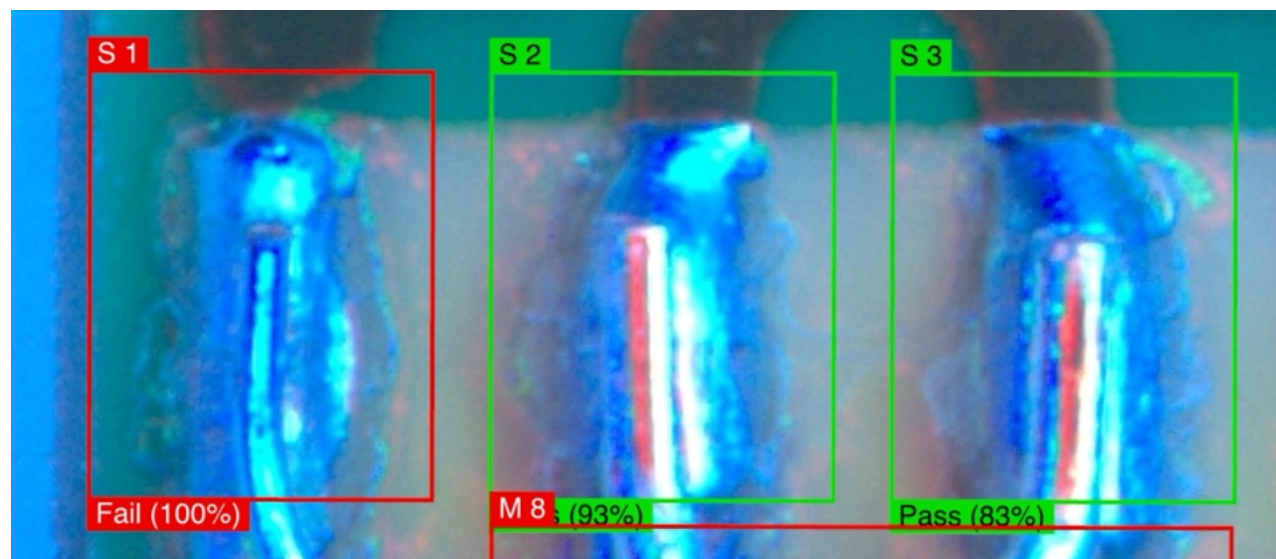
OV80i with advanced lighting and state-of-the-art AI catches most raised wire issues.



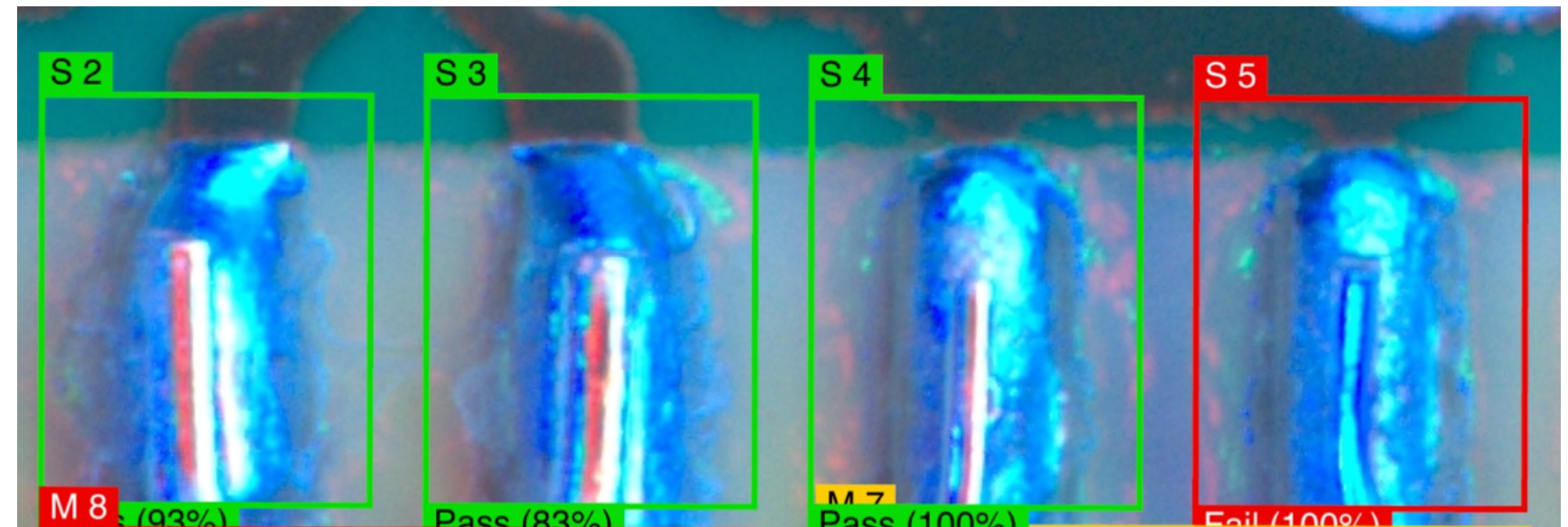
Very hard to see raised wire from top-down view (shown right)



Very hard to see raised wire from top-down view (shown right)

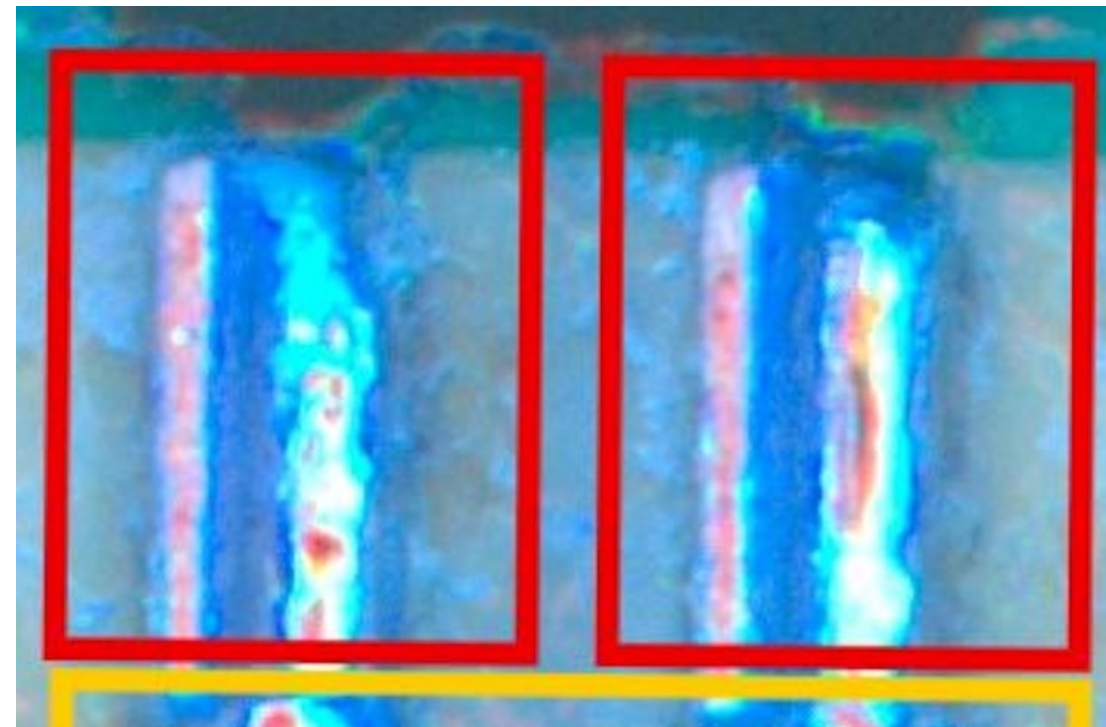
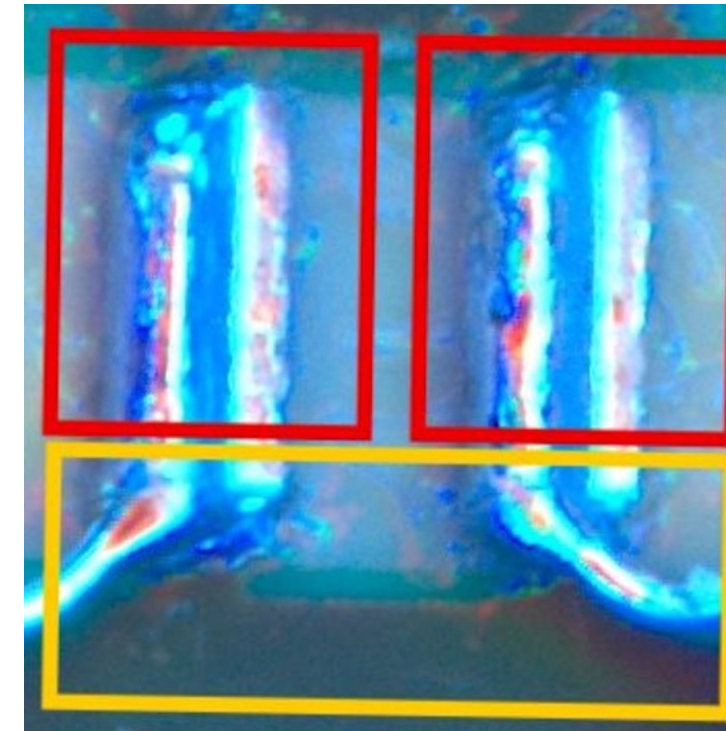
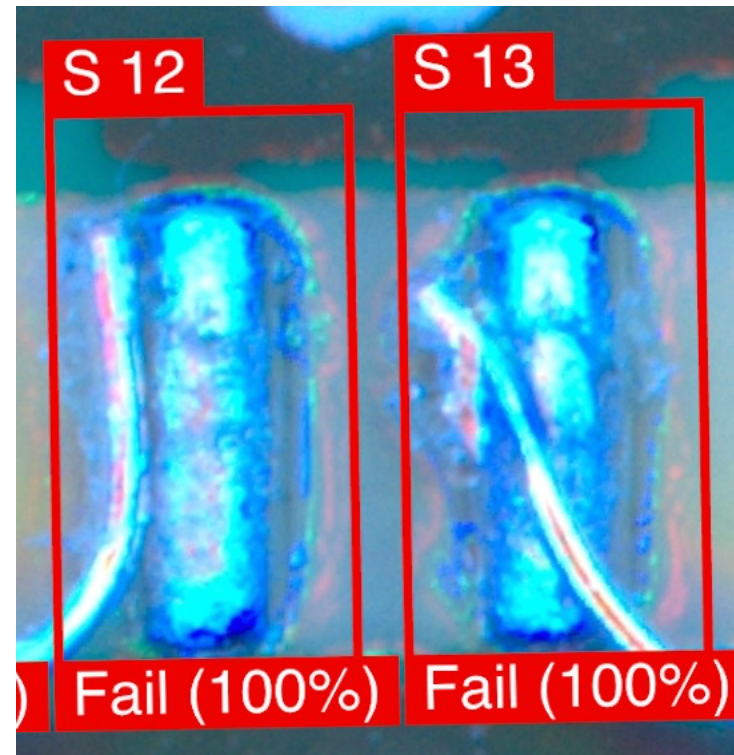


Special 2.5D lighting allows AI to detect this issue.



Special 2.5D lighting allows AI to detect this issue.

Example of Other Defect Types Caught





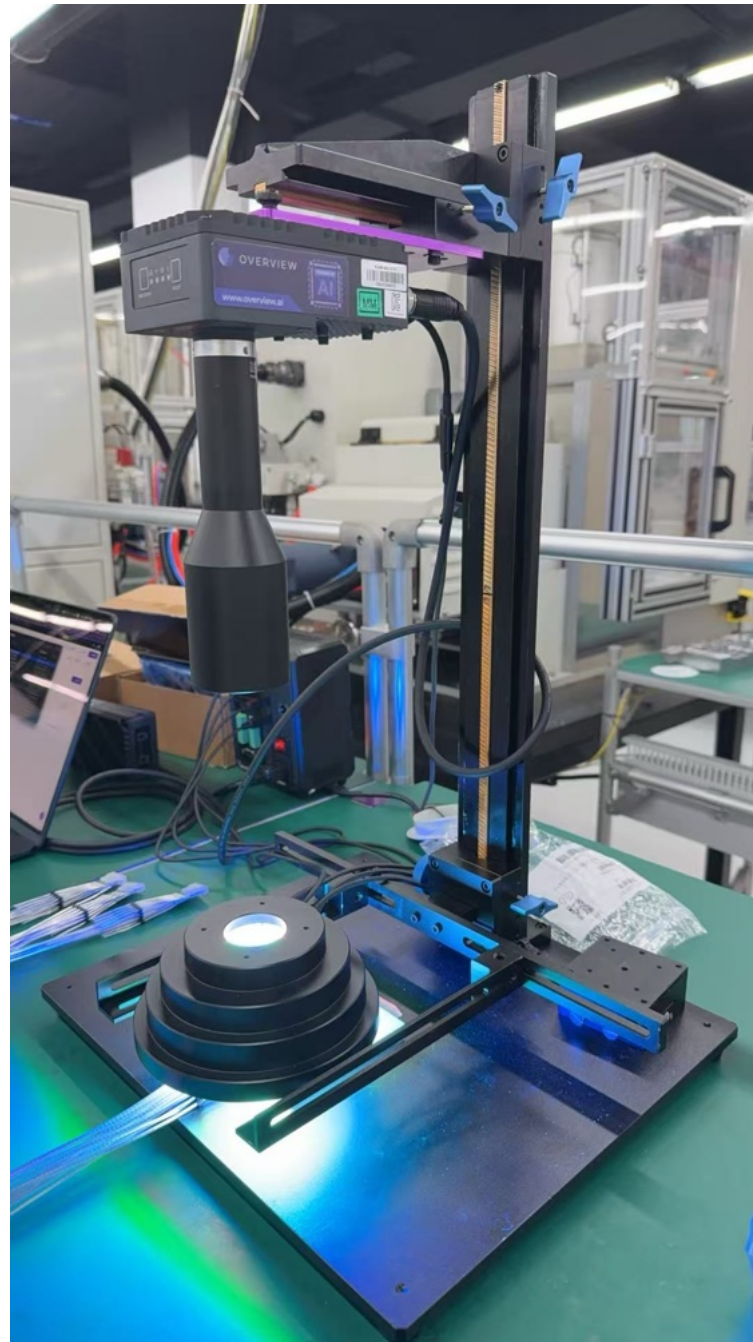
OVERVIEW

Configuring OV80i

Power of a vision system, ease-of-use of a vision sensor.

Step 1: Setup camera and lighting

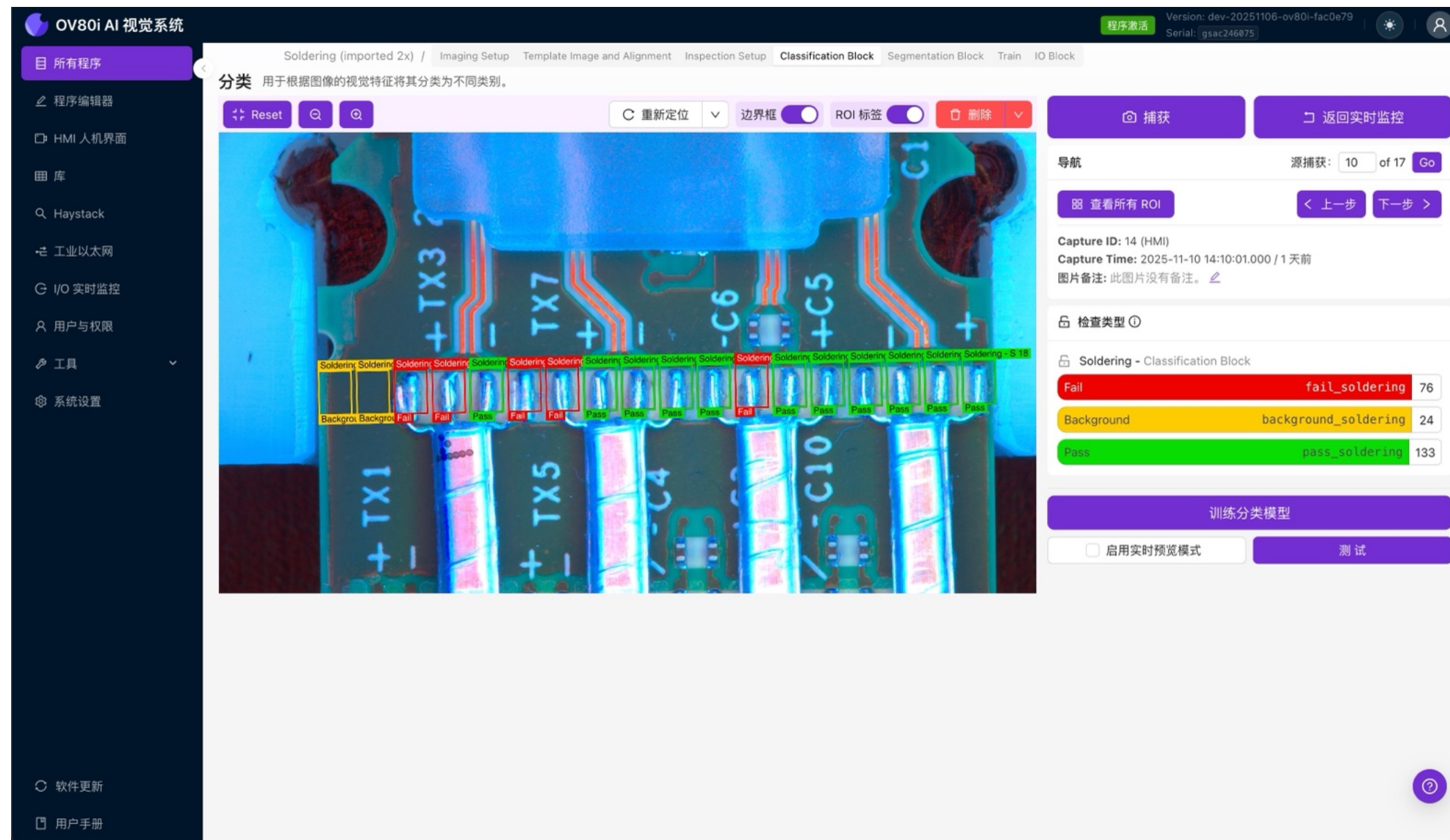
Optimizing image setup to increase accuracy and decrease difficulty of inspection.



1. Telecentric Lens 0.4X Zoom for image consistency.
2. 2.5D 3-Level Light to accentuate Z-axis issue (raised wire)

Step 2: Setup classifier inspection

Add dynamic inspection regions per solder on PCB with auto-alignment tool. Only trained on 17 images!



Easy setup: Simple labeling tools.

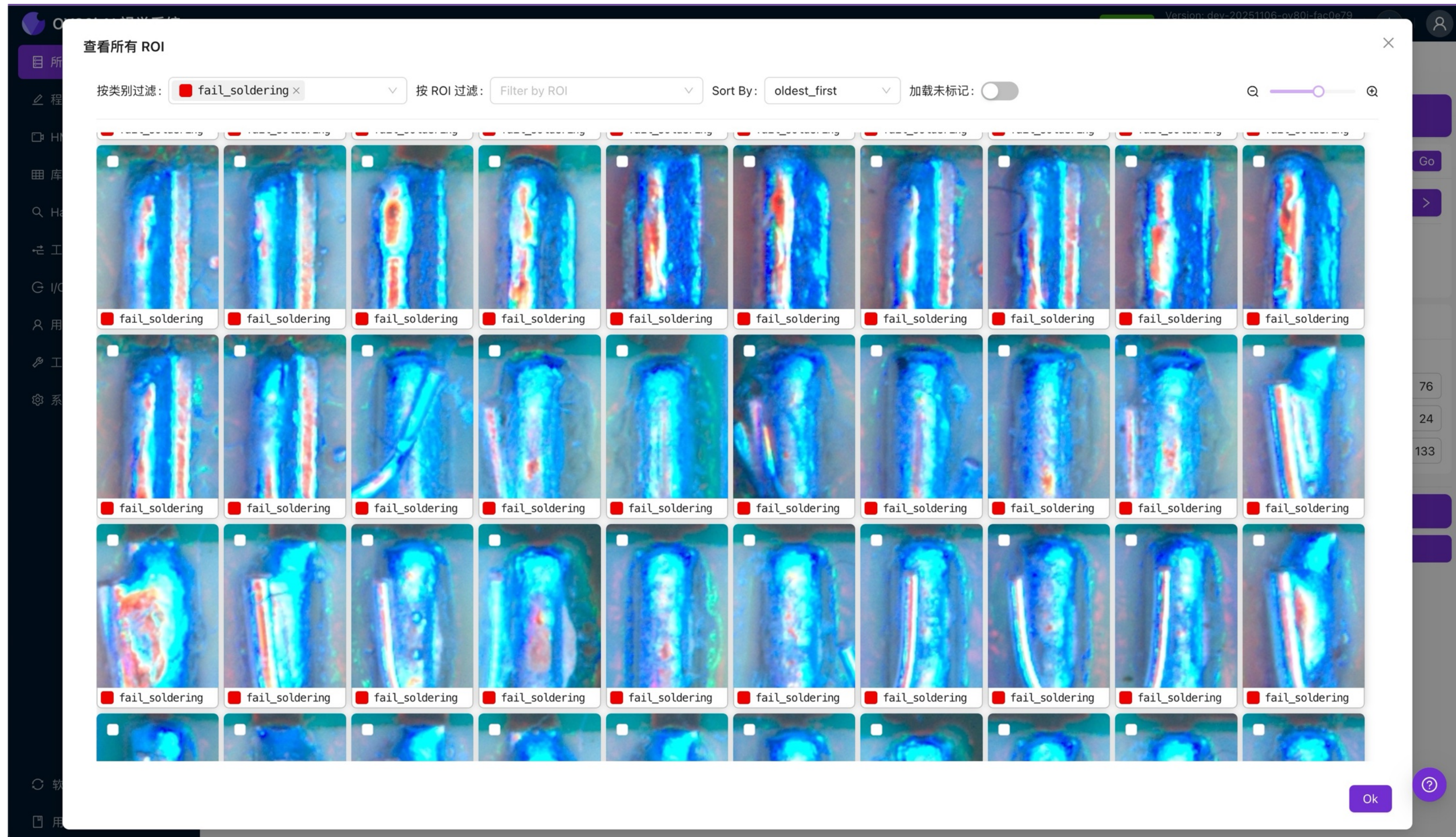
Multiple class options.

Step 3: Setup segmenter inspection

Layer on multi-class segmentation to find wire and check position + check for height and size of excess material. Only 9 images used!

The screenshot displays the OV80i AI Visual System interface. The top navigation bar includes tabs for Soldering (imported 2x), Imaging Setup, Template Image and Alignment, Inspection Setup, Classification Block, Segmentation Block (active), Train, and IO Block. The Segmentation Block is currently active, showing a live video feed of a soldering process. The video feed is overlaid with a segmentation mask, where different colors represent different classes. Four yellow bounding boxes are drawn on the video feed, labeled W1, W2, W3, and W4, indicating the positions of the wires. Below these, two more yellow bounding boxes are labeled M1 and M2, indicating the positions of the excess material. The interface also includes a left sidebar with various menu items, a top toolbar with buttons like Reset, 重新定位, 边界框, 显示注释, and 删除, and a right sidebar with buttons like Capture, Return to Live, and Save Annotations. The right sidebar also displays navigation information, including Source Capture (9 of 9), Capture ID (20), and Capture Time (2025-11-10 15:50:05.000).

Step 4: Check classification dataset



Step 5: Train model and track mislabels

Training happens locally on OV80i in under 5 minutes.

OV80i AI 视觉系统

所有程序

程序编辑器

HMI 人机界面

图库

Haystack

工业以太网

I/O 实时监控

用户与权限

系统设置

软件更新

用户手册

训练 AI 模型

分类 (Classification)

Training Model - 00:32 left 52%

✓ 训练准确性: 100% 总图像数: 231 迭代次数: 64 / 100

训练准确性

训练损失

正确

错误

训练

验证

✓	Crop	Result	Loss	Confidences	Class Name	Inspection Type	Type
☐		正确	0.910		Pass	Soldering	训练
☐		正确	0.238		Pass	Soldering	训练
☐		正确	0.202		Pass	Soldering	训练
☐		正确	0.260		Pass	Soldering	训练
☐		正确	3.105		Pass	Soldering	训练

64

中止训练

提前结束训练

实时预览

关闭

快速指南

导入捕获

总捕获: 18

上一步 下一步 >

Edit

ss_soldering 84

l_soldering 123

nd_soldering 24

类别 Background

24 张图像。这种类

产生不利影响。为

别添加更多图像,

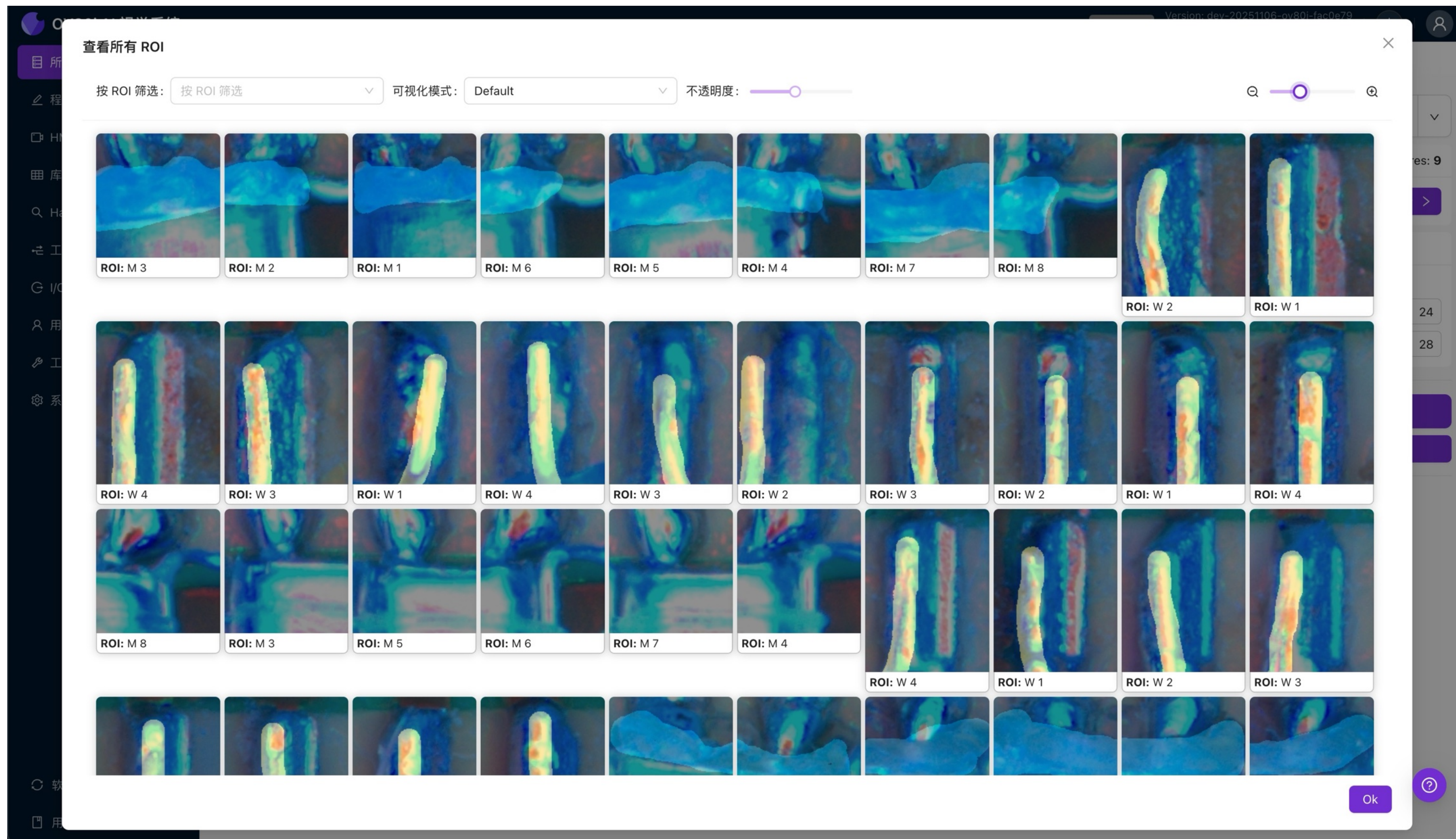
PM

训练分类模型

测试

?

Step 6: Check and train segmentation model



Step 7: Setup Data Output and Rule Builder

OV80i AI 视觉系统

程序激活

Version: dev-20251106-ov80i-fac0e79

Serial: gsac246075

所有程序

程序编辑器

HMI 人机界面

库

Haystack

工业以太网

I/O 实时监控

用户与权限

工具

系统设置

Soldering (imported 2x) / Imaging Setup Template Image and Alignment Inspection Setup Classification Block Segmentation Block Train IO Block

IO模块

Configure rules to define the inspection pass/fail logic.

Save & Deploy

Advanced Mode

Preview Test

Reset

Navigation

1 / 9

Prev

Next

Capture ID: 26 (Manual)

Capture Time: 2025-11-10 11:46:38.000 / 1 天前

图片备注: Cloned from recipe: Soldering (imported 2x) - Capture ID: 4

Classification Rules

(ID: 1) S 1 × (ID: 14) S 2 × (ID: 2) S 3 × (ID: 3) S 4 × (ID: 10) S 5 × (ID: 16) S 6 × (ID: 11) S 7 × (ID: 12) S 8 × (ID: 7) S 10 × (ID: 4) S 11 × (ID: 8) S 12 × (ID: 5) S 13 × (ID: 6) S 14 × (ID: 13) S 15 × (ID: 15) S 16 × (ID: 9) S 17 ×

≠

Fail

+

Add rule

Segmentation Rules

(ID: 19) M 1 × (ID: 20) M 2 × (ID: 21) M 3 × (ID: 22) M 4 × (ID: 23) M 5 × (ID: 24) M 6 × (ID: 25) M 7 × (ID: 26) M 8 ×

Material

Major Axis Length

All

<

50

+

Add rule

+

Add rule group

OVERVIEW

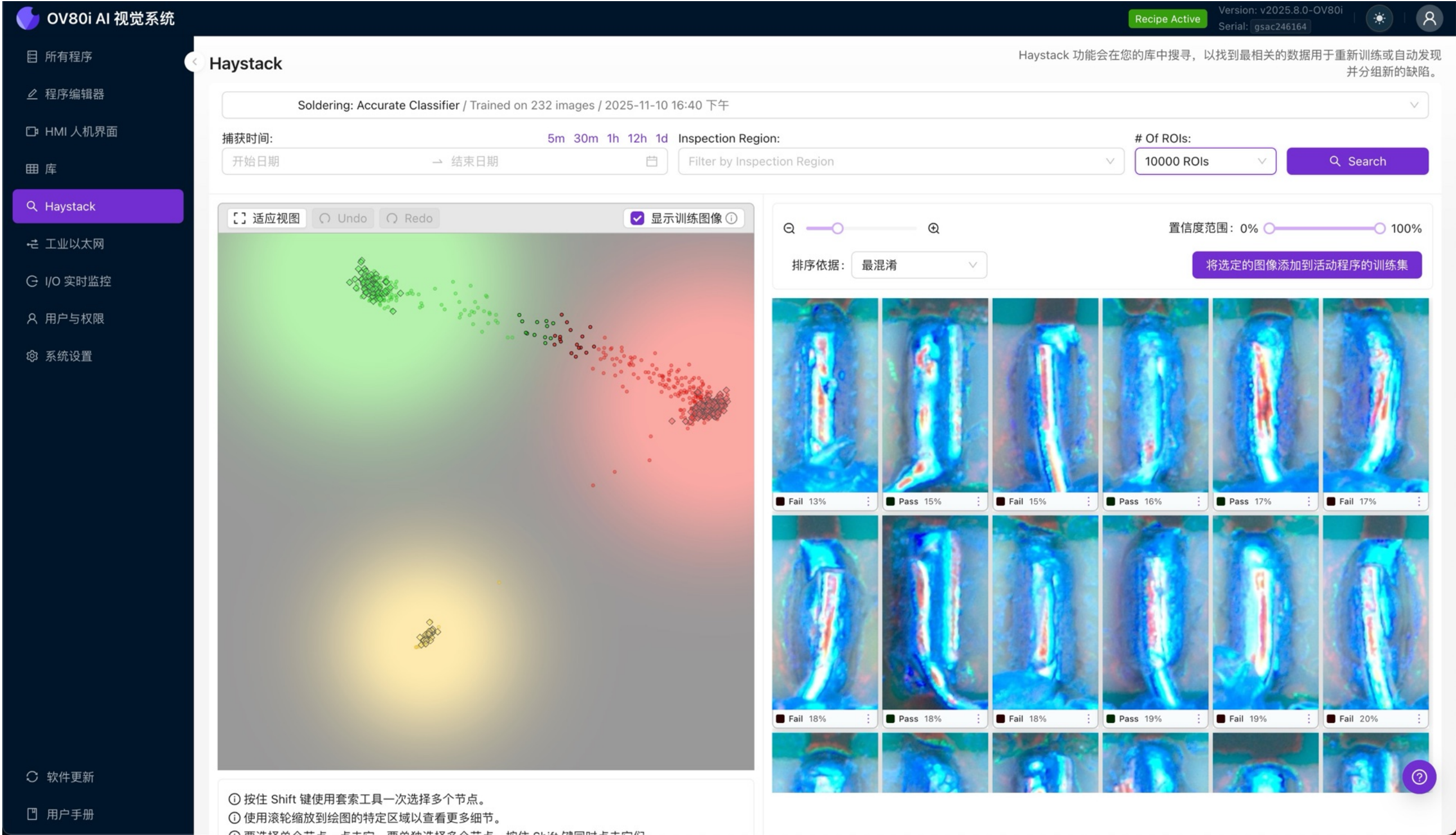


OVERVIEW

**Results: 1-2 hours to configure,
setup, label, train and deploy
each inspection model.**

OV80i Haystack

Analyze thousands of production images stored on device. Look for outliers, borderline cases and find anomalies and/or new defect types.



OV80i Library: Store 20,000 images on device.

Easier root cause analysis, faster retraining and redeploying. Easy production data retraining.

OV80i AI 视觉系统

所有程序

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软件更新

用户手册

库

#198

Soldering

2025-11-11 16:25:09.296

1天前

FAIL

#197

Soldering

2025-11-11 16:14:19.292

1天前

FAIL

#196

Soldering

2025-11-11 16:13:55.475

1天前

FAIL

#195

Soldering

2025-11-11 16:13:25.821

1天前

FAIL

#194

Soldering

2025-11-11 16:12:57.542

1天前

FAIL

#193

Soldering

2025-11-11 16:12:26.894

1天前

FAIL

#192

Soldering

2025-11-11 16:11:44.658

1天前

FAIL

#191

Soldering

2025-11-11 16:11:12.878

1天前

FAIL

#190

Soldering

2025-11-11 16:10:32.114

1天前

FAIL

#189

Soldering

2025-11-11 16:09:59.297

1天前

FAIL

#187

Soldering

2025-11-11 16:09:03.278

1天前

FAIL

#186

Soldering

2025-11-11 16:07:55.343

1天前

FAIL

Reset

边界框

ROI 标签

热图

箭头键 ↑↓ 可用于在图片之间移动。
按住 Shift 键可选择一系列图片。

Capture #198 from

Soldering

Trigger ID ①: 36 | Trigger Mode ①: 手动 HMI 触发模式

Capture Time: 2025-11-11 16:25:09.296 / 1天前

Process Time: 506 ms

⬇️ 下载 / ⬇️ 下载 (带标签) / 🔗 复制链接

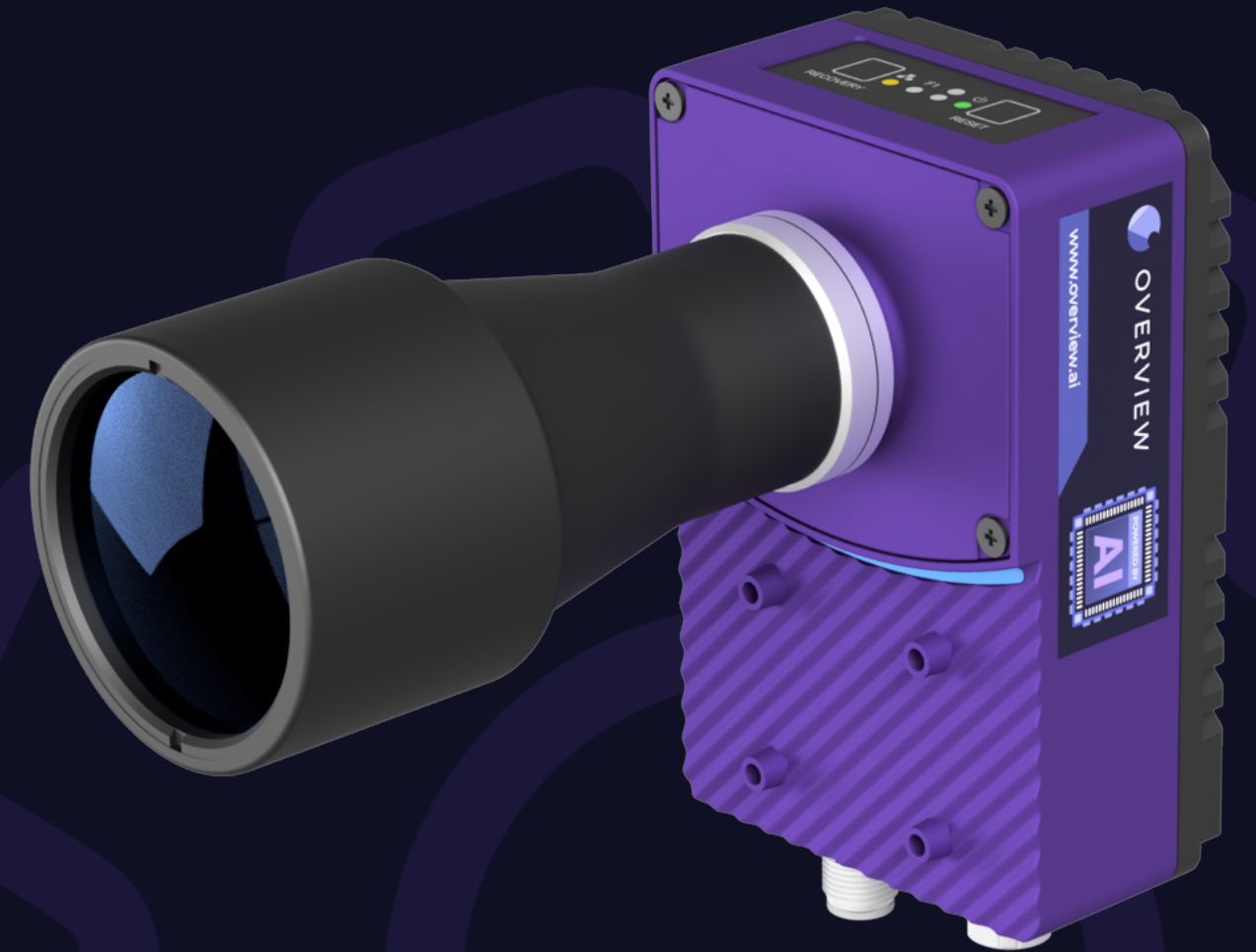
图片备注: 1 Over 🔗

Inspection	Confidence	Class
Material		4 Background 3 Fail 1 Pass
M 1	100%	Background
M 2	26.3%	Pass
M 3	99.9%	Background
M 4	100%	Fail
M 5	96.7%	Background

Custom notes stored in library by Engineer when double checking to verify inspection results.

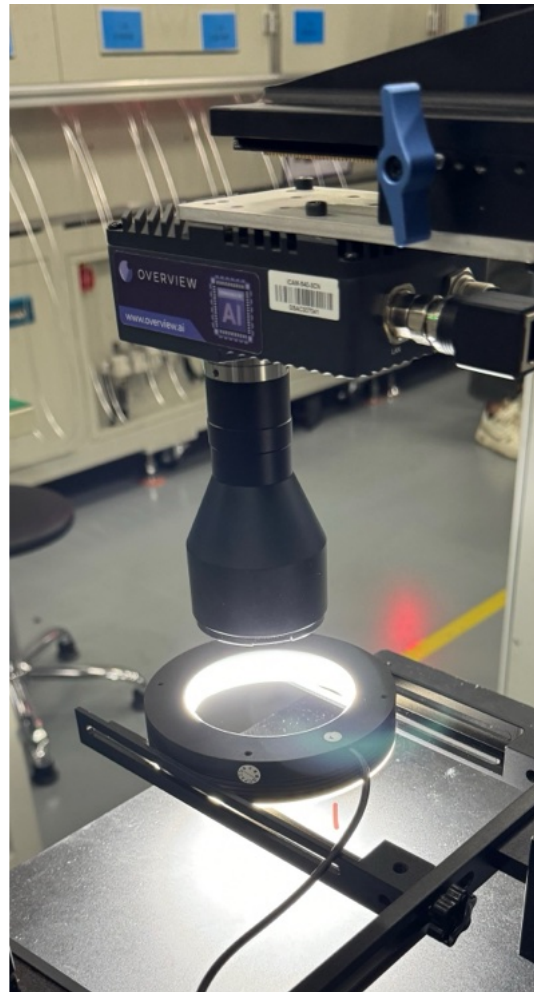
OV80i

AI Vision System

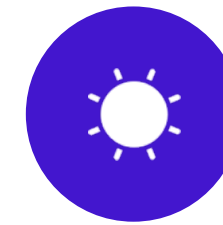


The OV80i

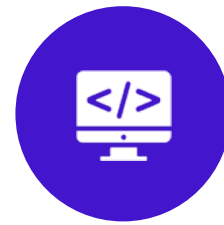
Designed for 10x Faster Installs. The truly all-in-one Deep Learning AI Vision System.



Imaging
8.3MP Color



On-board Library and
Root-Cause Analysis
Tool (Haystack)



Powerful Computing
- Onboard Nvidia GPU



No software download
needed

Industry Leading Accuracy. Built on reliable, proven hardware. No software downloads or special computer requirements. *Easy setup*, fast ramp-up for engineers and technicians. Rapid troubleshooting and maintenance.

No software licenses, no recurring fees.

OV80i Benefits and Advantages

1. Industry leading AI accuracy. Form-factor and ease-of-use of vision sensor.
2. Reduced variance in manual human inspection, AI inspection is 100% inspection, 24/7.
3. Reduce escapes of bad parts.
4. Onboard data. Simplify root cause analysis and eventual yield gains.
5. Easy to train – AI models can be trained in 30 minutes or less by manufacturing, process, controls or quality engineers.
6. Flexibility – OV80i can be used for small component defect detection or larger assembly verification tasks.
7. Fast inspections – as low as 0.3 seconds per image.
8. Visual AI Inspections that can help replace human inspectors.
9. Multi-language Support.



OVERVIEW

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🌐 www.overview.ai